

Chapter 7: Decimals

Three-week revision booklet · 6 worksheets · 180 questions

NSW syllabus outcome

MA4-FRC-C-01 — represents and operates with fractions, decimals and percentages · MAO-WM-01

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How to use this booklet

There are six worksheets, two for each week, and each one runs across **two pages**. Every sheet has **30 questions** split into **Easy**, **Medium** and **Challenging**, with room to show your working underneath each one. Each section opens with a **worked example** that takes you through the method step by step, plus a short reminder of the things students most often slip up on. Work from the top of the page down — the questions get harder as you go. The final worksheet mixes every skill together, so it is the best check that you are ready for the topic test.

Week	Worksheet	Skill
1	1	Ordering and rounding decimals
1	2	Decimals and fractions
2	3	Adding and subtracting decimals
2	4	Multiplying decimals
3	5	Dividing decimals
3	6	Mixed revision

Progress tracker

Worksheet	Date completed	Score / 30	Checked by
1 Ordering and rounding decimals			
2 Decimals and fractions			
3 Adding and subtracting decimals			
4 Multiplying decimals			
5 Dividing decimals			
6 Mixed revision			

Worksheet 1: Ordering and rounding decimals

Week 1 · Worksheet 1 of 6 · 30 questions

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Compare decimals digit by digit from the left. When rounding, look at the next digit along.

EASY

18 QUESTIONS

WORKED EXAMPLE

Which is bigger, 0.6 or 0.58? · Round 3.47 to 1 decimal place.

STEP BY STEP

- 1 Give both numbers the same number of digits: 0.6 becomes 0.60
- 2 Compare from the left. Tenths: **6** beats **5**, so **0.6 > 0.58**
- 3 To round 3.47 to 1 dp, cut after one digit: 3.4|7
- 4 The digit after the cut is 7, which is 5 or more, so round up: **3.5**

REMEMBER

- A longer decimal is not always bigger — 0.58 has more digits than 0.6 but is smaller.
- 5 or more, round up. Less than 5, leave it as it is.

Insert < or > to make each statement true.

1. 0.5 ___ 0.45

2. 3.2 ___ 3.19

3. 0.07 ___ 0.7

4. 0.9 ___ 0.89

5. 1.25 ___ 1.3

6. 8.1 ___ 8.09

7. 2.75 ___ 2.8

8. 0.04 ___ 0.004

9. 6.03 ___ 6.3

Round each number to one decimal place.

10. 4.68

11. 0.473

12. 12.05

13. 6.94

14. 3.14

15. 27.02

16. 1.44

17. 9.37

18. 0.28

Worksheet 1: Ordering and rounding decimals

Week 1 • Worksheet 1 of 6 • continued

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MEDIUM

8 QUESTIONS

WORKED EXAMPLE

Round 4.6749 to 2 decimal places.

STEP BY STEP

- 1 You need 2 digits after the point, so cut there: 4.67 | 49
- 2 Look only at the **first** digit after the cut, which is 4.
- 3 4 is less than 5, so the digit before the cut stays the same.
- 4 Answer: **4.67**

REMEMBER

- Ignore the rest of the tail — the 9 on the end does not matter.
- “To the nearest whole number” means cut straight after the point.

Round as directed.

19. 13.15 to 1 dp

20. 4.678 to 2 dp

21. 0.264 to 2 dp

22. 7.555 to 2 dp

23. 0.862 to 2 dp

24. 19.96 to 1 dp

25. 128.4 to the nearest whole number

26. 6.5 to the nearest whole number

CHALLENGING

4 QUESTIONS

WORKED EXAMPLE

Write in ascending order: 2.5, 2.05, 2.45

STEP BY STEP

- 1 Give them all the same number of decimal places: 2.50, 2.05, 2.45
- 2 They all start with 2, so compare the tenths: 5, 0 and 4.
- 3 Smallest tenths first: 0, then 4, then 5.
- 4 Answer: **2.05, 2.45, 2.5**

REMEMBER

- Ascending means smallest first; descending means largest first.
- Write the numbers in their **original** form in your answer.

Answer each question. Show your working.

27. Write in ascending order: 5.876, 5.99, 5.09, 5.8

28. Write in descending order: 3.021, 3.2, 3.19, 3.008

29. A length is 4.647 m. Give it to 2 dp and to 1 dp.

30. Explain why 0.30 and 0.3 are equal.

Worksheet 2: Decimals and fractions

Week 1 · Worksheet 2 of 6 · 30 questions

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A decimal is just a fraction with a denominator of 10, 100 or 1000.

EASY

18 QUESTIONS

WORKED EXAMPLE

Write 0.6 as a fraction. · Write $\frac{1}{4}$ as a decimal.

STEP BY STEP

- 1 0.6 has 1 decimal place, so it is 6 **tenths**: $0.6 = \frac{6}{10}$
- 2 Simplify by dividing top and bottom by 2: $\frac{6}{10} = \frac{3}{5}$
- 3 A fraction bar means “divide”, so $\frac{1}{4}$ means $1 \div 4$.
- 4 $1 \div 4 = 0.25$

REMEMBER

- 1 decimal place = tenths, 2 places = hundredths, 3 places = thousandths.
- Always simplify by dividing top and bottom by the **same** number.

Write each decimal as a fraction in simplest form.

1. 0.5

2. 0.2

3. 0.6

4. 0.25

5. 0.75

6. 0.4

7. 0.8

8. 0.35

9. 0.45

Write each fraction as a decimal.

10. $\frac{1}{2}$

11. $\frac{1}{4}$

12. $\frac{3}{4}$

13. $\frac{1}{5}$

14. $\frac{2}{5}$

15. $\frac{4}{5}$

16. $\frac{7}{10}$

17. $\frac{9}{10}$

18. $\frac{3}{10}$

Worksheet 2: Decimals and fractions

Week 1 • Worksheet 2 of 6 • continued

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● MEDIUM

8 QUESTIONS

WORKED EXAMPLE

Write 0.35 as a fraction. • Write $\frac{3}{8}$ as a decimal.

STEP BY STEP

- 1 0.35 has 2 decimal places, so it is 35 hundredths: $\frac{35}{100}$
- 2 Both 35 and 100 divide by 5: $\frac{35}{100} = \frac{7}{20}$
- 3 For $\frac{3}{8}$, divide the top by the bottom: $3 \div 8$
- 4 $3 \div 8 = 0.375$

REMEMBER

- Check your simplifying: is there still a number that divides both?
- Worth knowing: $\frac{1}{8} = 0.125$, $\frac{1}{4} = 0.25$, $\frac{1}{2} = 0.5$

Write each decimal as a fraction in simplest form.

19. 0.375

20. 0.45

21. 0.04

22. 0.125

Write each fraction as a decimal.

23. $\frac{5}{8}$

24. $\frac{11}{20}$

25. $\frac{3}{8}$

26. $\frac{17}{20}$

● CHALLENGING

4 QUESTIONS

WORKED EXAMPLE

Which is larger, $\frac{3}{5}$ or 0.55?

STEP BY STEP

- 1 You cannot compare a fraction and a decimal directly.
- 2 Change the fraction into a decimal: $3 \div 5 = 0.6$
- 3 Now give both the same number of places: 0.60 and 0.55
- 4 0.60 is larger, so $\frac{3}{5}$ is larger.

REMEMBER

- Put everything into the same form first — decimals are usually easiest.
- Show the converted value in your working, not just the final answer.

Answer each question. Show your working.

27. Order from smallest to largest: $\frac{3}{4}$, 0.7, 0.78

28. Which is larger, $\frac{5}{8}$ or 0.62? Show your working.

29. A pipe is 0.85 m long. Write this as a fraction of a metre in simplest form.

30. Write $\frac{9}{4}$ as a decimal and as a mixed numeral.

Worksheet 3: Adding and subtracting decimals

Week 2 · Worksheet 3 of 6 · 30 questions

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Line up the decimal points, fill any gaps with zeros, then add or subtract as usual.

EASY

18 QUESTIONS

WORKED EXAMPLE

Work out $12.6 + 3.45$

STEP BY STEP

- 1 Write them under each other with the decimal points lined up.
- 2 Fill the gap with a zero so both have 2 decimal places:
12.60
- 3 Add from the right, just like whole numbers: $12.60 + 3.45$
- 4 Bring the decimal point straight down: **16.05**

REMEMBER

- 12.6 and 12.60 are the same number — adding a zero changes nothing.
- The point in the answer sits directly under the points above it.

Evaluate each expression.

1. $12.6 + 3.4$

2. $8.2 + 1.9$

3. $5.75 + 2.25$

4. $0.6 + 0.45$

5. $14.3 + 7.8$

6. $9.05 + 3.6$

7. $2.4 + 6.75$

8. $3.62 + 4.38$

9. $6.4 + 0.85$

Evaluate each expression.

10. $9.6 - 3.2$

11. $15.05 - 7.8$

12. $4.3 - 1.75$

13. $10 - 4.6$

14. $7.25 - 3.5$

15. $5 - 0.85$

16. $31.4 - 9.09$

17. $84.3 - 27.6$

18. $12.5 - 4.25$

Worksheet 3: Adding and subtracting decimals

Week 2 • Worksheet 3 of 6 • continued

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● MEDIUM

8 QUESTIONS

WORKED EXAMPLE

Work out $20 - 6.4 - 2.85$

STEP BY STEP

- 1 Work from left to right, one step at a time.
- 2 Write 20 as 20.00 so there is something to borrow from.
- 3 $20.00 - 6.40 = 13.60$
- 4 $13.60 - 2.85 = 10.75$

REMEMBER

- A whole number can always be written with a point and zeros.
- Do not try to do both subtractions at once.

Evaluate each expression.

19. $8.5 + 12.75 + 3.2$

20. $0.75 + 15 + 2.4$

21. $20 - 6.4 - 2.85$

22. $13.3 + 0.82 + 5.6$

23. $9.9 - 3.45 + 1.2$

24. $50 - 12.75 - 8.5$

25. $18.4 - 9.65 + 3.2$

26. $100 - 37.42$

● CHALLENGING

4 QUESTIONS

WORKED EXAMPLE

A jug holds 2.5 L. 0.4 L is poured out. How much is left?

STEP BY STEP

- 1 "Poured out" means something is taken away, so subtract.
- 2 Write the number sentence: $2.5 - 0.4$
- 3 $2.5 - 0.4 = 2.1$
- 4 Answer in words and units: **2.1 L is left**

REMEMBER

- Decide add or subtract before you touch the numbers.
- Always finish with the correct unit (L, m, kg, \$).

Answer each question. Show your working.

27. Joe needs cable lengths 2.3 m, 1.9 m, 4.2 m and 3.8 m. How much altogether?

29. A bill of \$47.60 is paid with \$50. Find the change.

28. A jug holds 2.5 L and 0.85 L is poured out. How much remains?

30. A tank held 18.5 L. 4.75 L was used, then 2.3 L was added. How much is in it now?

Worksheet 4: Multiplying decimals

Week 2 · Worksheet 4 of 6 · 30 questions

NSW syllabus outcome

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Multiplying by 10, 100 or 1000 moves the digits to the left. Otherwise multiply as whole numbers, then count the decimal places.

EASY

18 QUESTIONS

WORKED EXAMPLE

Work out $4.36 \times 10 \cdot 2.6 \times 4$

STEP BY STEP

- 1 For $\times 10$, every digit moves **one place left**: 4.36 becomes **43.6**
- 2 For 2.6×4 , first ignore the decimal point: $26 \times 4 = 104$
- 3 Now count the decimal places in the question: 2.6 has **1**.
- 4 Put 1 decimal place back into the answer: **10.4**

REMEMBER

- $\times 10$ moves 1 place, $\times 100$ moves 2 places, $\times 1000$ moves 3 places.
- Check it looks sensible: 2.6×4 should be a little more than 10.

Evaluate each expression.

1. 4.36×10

2. 4.36×100

3. 0.7×10

4. 2.5×100

5. 0.08×10

6. 13.2×100

7. 6.7×10

8. 0.3×1000

9. 0.45×100

Evaluate each expression.

10. 2.6×4

11. 1.2×5

12. 0.7×8

13. 7.2×3

14. 6.4×2

15. 0.25×8

16. 5.5×0.6

17. 0.9×0.4

18. 3.5×4

Worksheet 4: Multiplying decimals

Week 2 • Worksheet 4 of 6 • continued

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MEDIUM

8 QUESTIONS

WORKED EXAMPLE

Work out 0.6×0.4

STEP BY STEP

- 1 Ignore both decimal points and multiply: $6 \times 4 = 24$
- 2 Count the decimal places: 0.6 has 1 and 0.4 has 1.
- 3 $1 + 1 = 2$ decimal places are needed in the answer.
- 4 So 24 becomes **0.24**

REMEMBER

- Multiplying by a number smaller than 1 makes the answer **smaller**.
- You may need to write zeros in front to fit all the places in.

Evaluate each expression.

19. 0.71×1.3

20. 4.8×2.5

21. 0.06×0.4

22. 12.5×0.8

23. 3.14×2

24. 0.35×40

25. $(0.3)^2$

26. 2.4×3.5

CHALLENGING

4 QUESTIONS

WORKED EXAMPLE

A shop sells 6 drinks at \$2.50 each. Find the total.

STEP BY STEP

- 1 "Each" tells you to multiply: total = $6 \times \$2.50$
- 2 Ignore the point: $6 \times 250 = 1500$
- 3 \$2.50 has 2 decimal places, so put 2 places back: 15.00
- 4 Answer: **\$15.00**

REMEMBER

- With money, always write two digits after the point.
- Estimate first: 6 lots of \$2.50 is about \$15, so this looks right.

Answer each question. Show your working.

27. A café sells 12 coffees at \$4.50 each. Find the total.

28. Petrol costs \$1.85 per litre. Find the cost of 40 L.

29. A rectangle is 3.5 m by 2.4 m. Find its area.

30. Explain why 0.5×0.5 is smaller than 0.5.

Worksheet 5: Dividing decimals

Week 3 · Worksheet 5 of 6 · 30 questions

NSW syllabus outcome

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Dividing by 10, 100 or 1000 moves the digits to the right. To divide by a decimal, multiply both numbers by 10 until the divisor is a whole number.

EASY

18 QUESTIONS

WORKED EXAMPLE

Work out $82.5 \div 10 \cdot 8.4 \div 4$

STEP BY STEP

- 1 For $\div 10$, every digit moves **one place right**: 82.5 becomes **8.25**
- 2 For $8.4 \div 4$, divide it as you would a whole number.
- 3 Keep the decimal point in the answer directly above the one inside.
- 4 $8.4 \div 4 = \mathbf{2.1}$

REMEMBER

- $\div 10$ moves 1 place, $\div 100$ moves 2 places, $\div 1000$ moves 3 places.
- Dividing by a whole number makes it smaller, so check the answer went down.

Evaluate each expression.

1. $82.5 \div 10$

2. $82.5 \div 100$

3. $6.4 \div 10$

4. $150 \div 100$

5. $0.9 \div 10$

6. $3.2 \div 100$

7. $0.56 \div 10$

8. $8 \div 100$

9. $4.5 \div 10$

Evaluate each expression.

10. $9.57 \div 3$

11. $8.4 \div 4$

12. $7.2 \div 6$

13. $0.96 \div 8$

14. $15.5 \div 5$

15. $2.4 \div 2$

16. $6.3 \div 9$

17. $13.2 \div 4$

18. $8.6 \div 2$

Worksheet 5: Dividing decimals

Week 3 • Worksheet 5 of 6 • continued

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MEDIUM

8 QUESTIONS

WORKED EXAMPLE

Work out $9.6 \div 0.4$

STEP BY STEP

- 1 You cannot divide by a decimal, so turn 0.4 into a whole number.
- 2 Multiply **both** numbers by 10: 9.6 becomes 96 and 0.4 becomes 4.
- 3 The question is now $96 \div 4$.
- 4 $96 \div 4 = 24$

REMEMBER

- Whatever you do to one number you must do to the other.
- Dividing by a decimal can make the answer **bigger** than you started with.

Evaluate each expression.

19. $9.6 \div 0.4$

20. $1.5 \div 0.5$

21. $4.8 \div 0.6$

22. $0.72 \div 0.9$

23. $0.36 \div 0.06$

24. $6 \div 0.25$

25. $12 \div 0.8$

26. $24.6 \div 0.2$

CHALLENGING

4 QUESTIONS

WORKED EXAMPLE

A 6 m ribbon is cut into 1.5 m pieces. How many pieces?

STEP BY STEP

- 1 "How many fit into" means divide: $6 \div 1.5$
- 2 Multiply both numbers by 10 to clear the decimal: $60 \div 15$
- 3 $60 \div 15 = 4$
- 4 Answer: **4 pieces**

REMEMBER

- Check by multiplying back: $4 \times 1.5 = 6$
- This answer counts pieces, so it should be a whole number.

Answer each question. Show your working.

27. A 0.5 L bottle is filled from a 7.5 L container. How many bottles?

29. A ribbon 4.8 m long is cut into 0.6 m pieces. How many pieces?

28. \$46.80 is shared equally between 8 people. How much each?

30. A rectangle has area 8.4 m^2 and width 2.4 m. Find its length.

Worksheet 6: Mixed revision

NAME _____

Week 3 · Worksheet 6 of 6 · 30 questions

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This sheet mixes every skill from the Decimals chapter. Work through it as though it were the topic test.

EASY

18 QUESTIONS

WORKED EXAMPLE

Decide which skill the question is testing, then use that method.

STEP BY STEP

- 1 + or -: line the decimal points up under each other.
- 2 × or ÷ by 10, 100, 1000: move the digits left or right.
- 3 **Other ×**: multiply without the points, then count the decimal places.
- 4 **Rounding**: cut in the right place, then look at the next digit along.

REMEMBER

- Read the whole question before you start writing.
- Estimate roughly what the answer should be so you can check it.

Evaluate each expression.

1. $12.6 + 3.45$

2. $9.6 - 3.2$

3. 4.36×10

4. $82.5 \div 10$

5. 2.6×4

6. $9.57 \div 3$

7. $0.6 + 0.45$

8. $5 - 0.85$

9. 4.36×100

10. $82.5 \div 100$

11. 0.7×8

12. $0.96 \div 8$

Round each number to one decimal place.

13. 0.473

14. 4.68

15. 12.05

16. 6.94

17. 27.02

18. 1.44

Worksheet 6: Mixed revision

Week 3 • Worksheet 6 of 6 • continued

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● MEDIUM

8 QUESTIONS

WORKED EXAMPLE

Work out $0.75 + 15 + 2.4$

STEP BY STEP

- 1 Give every number the same number of decimal places.
- 2 0.75 stays as it is, 15 becomes 15.00 and 2.4 becomes 2.40.
- 3 Line the points up and add: $0.75 + 15.00 + 2.40$
- 4 Answer: **18.15**

REMEMBER

- A whole number can always be given a point and zeros.
- Estimate: $0.75 + 15 + 2.4$ is about 18, so 18.15 looks right.

Write each decimal as a fraction, or each fraction as a decimal.

19. 0.6

20. 0.25

21. $\frac{3}{4}$

22. $\frac{2}{5}$

Evaluate, or round as directed.

23. 0.71×1.3

24. $9.6 \div 0.4$

25. $20 - 6.4 - 2.85$

26. 13.15 to 1 dp

● CHALLENGING

4 QUESTIONS

WORKED EXAMPLE

A rectangle is 4 m by 2.5 m. Find its area.

STEP BY STEP

- 1 Write the rule you need: area = length $\times$ width
- 2 Put the numbers in: area = 4×2.5
- 3 $4 \times 25 = 100$, and 2.5 has 1 decimal place, so the answer is 10.
- 4 Answer: **10 m²**

REMEMBER

- Area is measured in **square** units, so write m².
- Write the rule down first — it earns marks even if you slip up later.

Answer each question. Show your working.

27. Write in ascending order: 5.876, 5.99, 5.09, 5.8

28. Joe needs cable lengths 2.3 m, 1.9 m, 4.2 m and 3.8 m. How much altogether?

29. A rectangle is 3.5 m by 2.4 m. Find its area.

30. \$46.80 is shared equally between 8 people. How much each?
